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ORIGINAL RESEARCH ARTICLE

Cost Effectiveness of a Pharmacist-Led Information Technology Intervention for Reducing Rates of Clinically Important Errors in Medicines Management in General Practices (PINCER)

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Abstract
Background and Objective We recently showed that a pharmacist-led information technology-based intervention (PINCER) was significantly more effective in reducing medication errors in general practices than providing simple feedback on errors, with cost per error avoided at £79 (US$131). We aimed to estimate cost effectiveness of the PINCER intervention by combining effectiveness in error reduction and intervention costs with the effect of the individual errors on patient outcomes and healthcare costs, to estimate the effect on costs and QALYs.
Methods We developed Markov models for each of six medication errors targeted by PINCER. Clinical event probability, treatment pathway, resource use and costs were extracted from literature and costing tariffs. A composite probabilistic model combined patient-level error models with practice-level error rates and intervention costs from the trial. Cost per extra QALY and cost-effectiveness acceptability curves were generated from the perspective of NHS England, with a 5-year time horizon.
Results The PINCER intervention generated £2,679 less cost and 0.81 more QALYs per practice [incremental cost-effectiveness ratio (ICER): −£3,037 per QALY] in the deterministic analysis. In the probabilistic analysis, PINCER generated 0.001 extra QALYs per practice compared with simple feedback, at £4.20 less per practice. Despite this extremely small set of differences in costs and outcomes, PINCER dominated simple feedback with a mean ICER of −£3,936 (standard error £2,970). At a ceiling 'willingness-to-pay' of £20,000/QALY, PINCER reaches 59 % probability of being cost effective.
Conclusions PINCER produced marginal health gain at slightly reduced overall cost. Results are uncertain due to the poor quality of data to inform the effect of avoiding errors.

On behalf of the PINCER Team.

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Key Points for Decision Makers

A pharmacist-led intervention to reduce prescribing and monitoring errors in primary care produced marginal health gain at slightly reduced overall cost.

The poor quality of data to inform the effect of avoiding errors introduced a lot of uncertainty into the analysis, and contributed significantly to the equivocal findings.

Targeting NSAID prescribing and amiodarone monitoring errors were the most cost-effective activities.

Targeting errors with evidence of effects on patient outcomes could prove more cost effective than targeting all errors, and should be considered by policy makers.

1 Introduction

Most health systems are attempting to improve patient safety [1, 2], and medication errors are considered an important cause of avoidable morbidity and mortality [3, 4]. Adverse events in England may be responsible for 850,000 inpatient episodes, costing £2 billion in bed-days, and increased mortality [3, 5]. In NHS England, the recent White Paper, *Equity and Excellence: Liberating the NHS*, refers specifically to strategies to improve patient safety, including prescribing [6].

Strategies to change prescribing behaviour are generally costly, with little evidence of cost effectiveness [7–9]. Furthermore, prescribing behaviour may not change as anticipated, and/or the clinical and resulting economic effects of most errors may be minor [10]. There have been few attempts to examine medical errors and adverse events from an economic perspective, far less to examine the cost effectiveness of interventions to reduce errors [11]. In a financially constrained healthcare environment, it is essential to be clearer about the true economic impact of medication error reduction.

Systematic reviews demonstrate the high incidence of prescription errors in primary care, the errors causing most harm and where these errors occur in the medicines use process [12, 13]. There is little evidence that describes the clinical and economic impact of medication error reduction [1]. Studies reporting interventions to reduce errors may provide information around costs of the intervention or effects on prescribing budgets [14], but not evidence around the effect of the intervention on patient outcome or costs.

Our cluster randomised controlled trial (RCT) demonstrated that a pharmacist-led information technology-based intervention (PINCER) reduced medication errors in general practices by 12.7 errors per practice at a cost per error avoided of £66 in year 2010 values (£79 in year 2012 values) [15]. This economic analysis only included PINCER intervention costs. Not knowing the impact of the intervention on patient health and NHS costs is an important limitation as the current UK evaluative framework requires a cost per QALY to compare the value for money of different healthcare interventions. Therefore, the aim of the economic analysis in this paper differs from our earlier preliminary analysis in that patient outcomes and healthcare costs incurred are estimated, rather than relying on the intermediate process indicator of errors.

2 Methods

2.1 Study Design, Comparators and Outcomes

Following published economic evaluation reporting criteria [16], we compared the costs and patient outcomes of the PINCER intervention with simple feedback in reducing rates of six clinically important medication errors in 72 English general practices. Simple feedback to practices comprised information from their computer systems on patients at risk of medication error and educational material on each error. The PINCER intervention involved—in addition to simple feedback—a pharmacist working within practices for a 12-week period to correct existing errors and put in place measures to prevent future problems. Data were extracted at baseline and at 6 and 12 months post-intervention. Primary [NSAIDs, β-blockers (β-adrenoceptor antagonists), ACE inhibitors (ACEIs)] and secondary (methotrexate, lithium, amiodarone) outcome measures were chosen as errors considered important in terms of overall burden and severity of iatrogenic harm in primary care [17–19], and those detectable from general practice computer systems (Table 1). Rationale for the choice of each measure is given in the trial protocol [20]. The PINCER trial sample size was determined according to the primary outcome measures at 6 months. Full details of the trial methods, results and initial economic analysis are available in the Electronic Supplementary Material and further detail is available elsewhere [15, 21].

The cost per extra QALY generated was determined. We took the perspective of the funder (NHS England) in terms of direct costs of providing the intervention and managing the consequences of errors. This analysis combined the results from the PINCER trial with literature-derived error-specific projected harm to generate estimates of patient outcomes and NHS costs (Fig. 1).

Table 1 Practice characteristics, prevalence of prescribing and monitoring problems at 6 months' follow-up, and intervention costs by treatment arm

	Simple feedback	PINCER	
Number of practices (%)	36 (50.0)	36 (50.0)	
Median Index of Multiple Deprivation 2004 score (IQR)	26.3 (18.8, 36.5)	30.3 (18.2, 39.6)	
Median list size (IQR)	6,438 (3,834, 9,707)	6,295 (2,911, 9,390)	
GP training practices (%)	10 (27.8)	13 (36.1)	
Total intervention cost per practice [£; year 2012 values] (95 % CI)[a]	98 (NA)	1,113 (349–2,212)	
Outcome/population at risk	Number of errors/number of patients at risk at 6 months		Relative risk reduction[a]
Primary outcomes			
NSAID: patients with a history of peptic ulcer prescribed an NSAID without a PPI/patients with a history of peptic ulcer without a PPI[b] (%)	86/2,014 (4.3)	51/1,852 (2.8)	0.35 $p = 0.01$
β-blocker: patients with asthma prescribed a β-blocker/patients with asthma[b] (%)	658/22,224 (3.0)	499/20,312 (2.5)	0.17 $p = 0.006$
ACEI: patients aged ≥75 on long-term ACEIs or diuretics without urea and electrolyte monitoring in the previous 15 months/patients aged ≥75 on long-term ACEIs or diuretics[c] (%)	436/5,329 (8.2)	255/4,851 (5.3)	0.36 $p = 0.003$
Secondary outcomes			
Methotrexate: patients prescribed methotrexate for ≥3 months without a full blood count or liver function test in last 3 months/patients prescribed methotrexate for ≥3 months[c] (%)	162/518 (31.3)	122/494 (24.7)	0.19 $p = 0.45$
Lithium: patients prescribed lithium for ≥3 months without a lithium level in last 3 months/patients prescribed lithium for ≥3 months[c] (%)	84/211 (39.8)	67/190 (35.3)	0.11 $p = 0.12$
Amiodarone: patients prescribed amiodarone for ≥6 months without a thyroid function test in the last 6 months/patients prescribed amiodarone for ≥6 months[c] (%)	106/235 (45.1)	81/242 (33.5)	0.25 $p = 0.02$

ACEI ACE inhibitor, *GP* general practice, *IQR* interquartile range, *NA* not applicable, *PPI* proton pump inhibitor

[a] Corrected for practice size [15]

[b] Prescribing error

[c] Monitoring error

2.2 Intervention Costs

The costs associated with simple feedback arose from researchers visiting practices at set time periods to generate error reports, reflecting the equivalent resource that would be consumed in practices to generate these reports. The PINCER intervention comprised these costs, plus pharmacist training sessions, facilitated meetings, monthly meetings, practice feedback meetings and time spent in each practice outside meetings following up errors [15, 20].

Three reports were run in each practice (at baseline and at 6 and 12 months' follow-up), costing £98.41 per practice at 6 months' follow-up, and £147.62 per practice at 12 months' follow-up. In total, report generation cost £3,542.77 and £5,314.16 for 36 simple feedback and 36 PINCER intervention practices at 6 and 12 months, respectively. The PINCER arm also generated £10,529.26 training, £109.07 preparation, £7,395.42 facilitated meeting, £2,116.08 monthly meeting, £842.19 practice meeting and £15,519.67 error management costs. The components were summed together to give the total mean cost per practice in each arm (Table 2). Further detail on methods is in the Electronic Supplementary Material.

2.3 Clinical and Economic Impact of Errors

PINCER reduced error rates across multiple disease areas. Most economic analyses operate within one therapeutic area. To generate costs and QALYs for PINCER we used a novel approach where economic models were developed for each error.

We modelled the effect of the observed error reductions at 6 months on patient outcomes and NHS costs, reflecting the timeframe for the primary outcomes. The outcomes described in Table 1 were included, requiring development of six models describing the consequences of the errors (see Fig. 1).

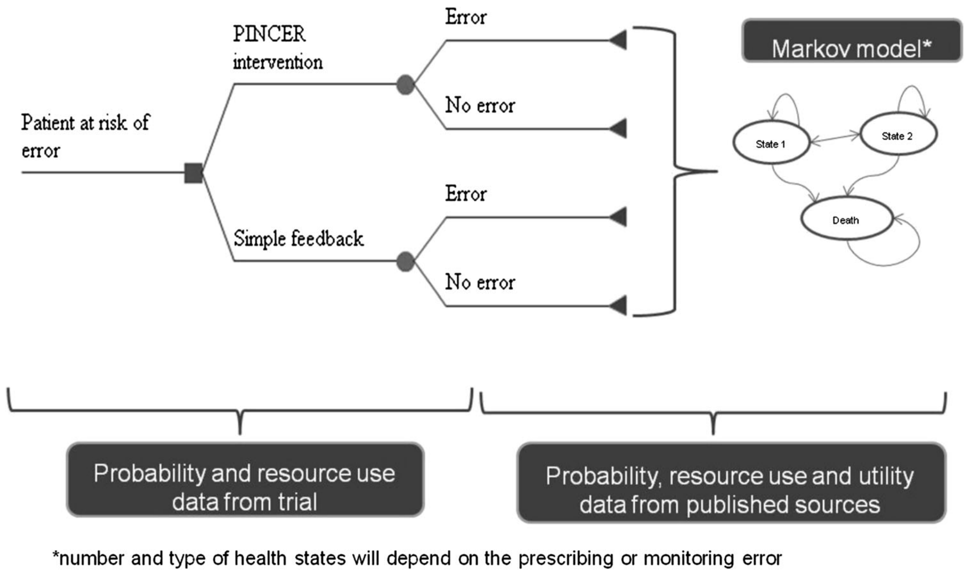

Fig. 1 Overview of economic model developed to combine PINCER trial results with estimates of harm caused by errors

Table 2 Summary of costs, outcomes and incremental economic analyses associated with PINCER intervention and simple feedback (costs inflated from year 2010 to 2012 values) [53]

Mean cost per practice (median, range)[a] [£; year 2012 values]	Simple feedback	PINCER
Report generation		
6 months	98.41	98.41
12 months	147.62	147.62
Pharmacist training costs	0	292.48 (283.83, 84.31–626.70)
Quarterly facilitated strategic meetings	0	206.94 (200.82, 59.66–443.43)
Monthly operational meetings	0	60.29 (58.51, 17.38–129.19)
Practice feedback	0	23.39 (22.71, 6.74–50.13)
Management of errors	0	431.10 (340.19, 60.46–1,397.80)
Total cost		
6 months	98.41	1,112.65 (1,025.93, 348.97–2,211.99)
12 months	147.62	1,161.86 (1,075.14, 398.18–2,261.19)
Mean incremental cost (£) [SD (95 % CI)]		
6 months		1,014.24 (908.32–1,120.16)
12 months		1,014.25 (910.24–1,118.24)
Mean incremental errors [SD (95 % CI)]		
6 months		12.90 (0.26, 13.42–12.39)
12 months		12.71 (0.29, 13.27–12.14)
Mean ICER per error avoided [£] (2.5–97.5th percentile)		
6 months		78.62 (70.41–86.83)
12 months		79.86 (71.67–88.05)

ICER incremental cost-effectiveness ratio, *SD* standard deviation

[a] Unless stated otherwise

Markov models were designed and populated with probabilistic data. Each model had a 3-month cycle length with half-cycle correction, a 5-year time horizon, 2012 cost year, and the UK Treasury-recommended 3.5 % discount rate for costs and outcomes [22]. Summaries of model development are available in the Electronic Supplementary Material. A full account is available elsewhere [23].

2.4 Sources of Data

In the NSAIDs model, with some drugs in this class being quite recently licensed and existing economic models focusing on relative gastrointestinal safety, we were able to build on existing search strategies and access relevant evidence for probabilities, costs and utilities, and use published economic models to optimise design. In the other five models, the drugs involved are established therapies available as generics with few or no economic models having been built for regulatory purposes or to investigate safety, so the quantity and quality of evidence was much older and poorer, requiring more extensive searching, consultation with clinical experts and tentative model design.

2.5 Literature Searches

The literature search was conducted through MEDLINE, EMBASE and Web of Science search terms that were specific to the errors (see the Electronic Supplementary Material). References in English and limited to humans were included. Databases were searched to the end of 2010. In the NSAIDs model we built on existing search strategies and a published systematic review and economic evaluation [24, 25]. No existing search strategies were available for other models and, despite including broad inclusion criteria, searches yielded low quantities of evidence. Comprehensive hand searches of the literature were also employed but in some instances only poor-quality published evidence was identified.

2.6 Individual Model Design

Model design was informed by published models where possible. In NSAIDs, there are many models that address gastrointestinal toxicity [25, 26]. For the β-blocker model, we adapted published models of asthma management [27, 28]. For outcomes where no published models exist, we had to develop new models with advice from clinical experts (see Acknowledgments). This included a model examining the effect of lack of monitoring of ACEIs on potassium levels or renal function or the effect on patient outcomes; a model examining the effect of lack of monitoring in methotrexate; a lithium model incorporating lithium intoxication where the literature is complex and contradictory [29–31]; and a model examining the effect of lack of amiodarone monitoring, where there is lack of consensus around management [32].

2.7 Transition Probabilities

Data came preferentially from up-to-date UK sources that reflected the PINCER trial patient characteristics. The amount, quality and relevance of evidence varied greatly for the different models. We accessed data from relevant RCTs and meta-analyses for the NSAIDs model to provide relatively reliable estimates of probabilities in the appropriate patient group. In other models, such as the β-blocker and ACEI models, there was very little relevant published evidence, despite very inclusive search strategies, requiring discussion with clinical experts. Definitions for outcomes varied in the literature, such as hyperkalaemia (ACEI model) [33, 34] and relapse in bipolar disorder (lithium model) [35, 36].

2.8 Identifying the Effect of the Error on Outcomes

Identifying the effect of errors on outcomes was straightforward in the NSAIDs model, as risk reductions were documented for patients who were prescribed an NSAID with a proton pump inhibitor versus patients with NSAIDs alone in the study by Pettit et al. [37]. Observational data then provided relevant, detailed and up-to-date information on death rates associated with hospitalisation for a gastric bleed in the UK [24]. However, in other models little evidence was found around the effect of the error, requiring the use of multiple sources of data (methotrexate model) or expert opinion (ACEI model). In the ACEI model, probabilities were available for developing acute renal failure (ARF) in the presence of [38] and absence of [39, 40] monitoring and death due to ARF [41]. However, the probability of developing hyperkalaemia as a result of ACEIs was not documented and, after discussion with clinical experts, we assumed a similar rate as for development of ARF in the absence of monitoring. In the amiodarone model, no studies reported the effects of monitoring thyroid function on patient outcomes. If patients are monitored, it is assumed that they will have a lower probability of staying in the untreated states with the associated increased risk of morbidity and mortality. If a patient is being monitored regularly, thyrotoxicity will be picked up and treated within one cycle. The probability of surgical management via thyroidectomy in amiodarone-induced thyrotoxicity is 0.081 [42]. It was assumed that the probability will be higher than zero as patients may be picked up by chance, at a rate of 10 % of the rate in the monitored group.

2.9 Health Status

In the NSAIDs model, directly relevant health status valuation was available for each of the health states [43] and this was also available in the lithium model for non-depressive health states [44]. In other models (β-blockers, ACEIs, lithium), we had to approximate relevant health state valuations from the literature, e.g. by using data from patients with major depression to approximate for bipolar patients in a depressive relapse [45]. In the methotrexate model, we had to estimate health states for bone marrow suppression due to the lack of available published estimates.

2.10 Resource Use and Unit Costs

Resource use data came preferentially from up-to-date UK sources of observation of clinical practice, with disaggregated resource use data, to allow attachment of current unit prices. If possible, individual patient data were used, with associated measures of mean and variation. If these were not available, point estimates were used, with carefully specified deterministic ranges, and standard methods for allocating distributions to these data were used. In the NSAIDs model, previous work by Elliott and colleagues [25] provided disaggregated resource use data for each health state. In the β-blocker and lithium models, we utilised published data [28, 46]. However, in other models, such as the ACEI and methotrexate model, there was very little published evidence, requiring reliance on clinical experts.

The probability, cost and utility data were assigned beta, gamma and beta distributions, respectively, and are summarised in Tables 3 and 4. Costs are given in year 2012 values.

2.11 Incremental Analysis

Each model was populated with probability, cost and health status data. This allowed the generation of the outcomes and costs in a cohort with the error present, and in a cohort with the error absent.

PINCER is a practice-level intervention with an associated practice-level effect and intervention costs. The error rates are reported at practice level but the Markov models generate costs and outcomes at patient level. Therefore, we needed to estimate, per practice, the difference in costs and outcomes generated for the PINCER and simple feedback arms. For the composite economic model, we assumed a practice with the characteristics of the mean in the PINCER trial practices. In the 36 practices in the control arm, at baseline, there were 28,769 patients in total at risk of one of the six errors [NSAIDs: 1,970 (7 %); β-blockers: 20,634 (71 %); ACEIs: 4,722 (16 %); methotrexate: 966 (4 %); lithium: 224 (1 %); amiodarone: 253 (1 %)]. This means that, in one typical practice in the PINCER trial, 799 patients were at risk of one of the six errors (NSAIDs: 55; β-blockers: 573; ACEIs: 131; methotrexate: 27; lithium: 6; amiodarone: 7). Therefore, in the composite economic model we estimated the difference in error rate between PINCER and simple feedback interventions at 6 months' follow-up for this practice population. The incidence rate for each error at 6 months' follow-up in the PINCER and simple feedback arms were combined with the appropriate error-specific model. Using the error-specific models, we generated the difference in patient outcome and costs between PINCER and simple feedback for each error for this practice population. Probabilistic estimates of costs and outcomes were derived, the analysis generating 5,000 iterations, using Monte-Carlo simulation for each error. The model assumes that no new patients enter the practice during the 5-year period.

This allowed us to generate the incremental impact of the PINCER intervention costs and outcomes for each error group in the practice population. The incremental costs and outcomes associated with each error were incorporated additively into the economic model. This allowed derivation of the total incremental impact of the PINCER intervention costs and outcomes for all six errors for one practice. At this point, the practice-level PINCER intervention or simple feedback costs were combined with these data.

Deterministic and probabilistic incremental economic analyses were carried out using these cost and outcome data. The incremental cost per extra QALY generated by PINCER over simple feedback was calculated using the following equation:

$$(\mathrm{Cost}_{\mathrm{PINCER}} - \mathrm{Cost}_{\mathrm{Simple\ feedback}}) / (\mathrm{QALY}_{\mathrm{PINCER}} - \mathrm{QALY}_{\mathrm{Simple\ feedback}})$$

It is not possible to generate 95 % confidence intervals around incremental cost-effectiveness ratios (ICERs) because the ratio of two distributions does not necessarily have a finite mean or, therefore, a finite variance [47]. Therefore, generation of a bootstrap estimate of the ICER sampling distribution to identify the magnitude of uncertainty around the ICERs is required. Bootstrapping with replacement was employed, utilising Microsoft® Excel®, using a minimum of 5,000 iterations to obtain 2.5 and 97.5 % percentiles of the ICER distribution.

Negative ICERs are difficult to interpret and often arise when one of the interventions is either 'dominant' (more effective, less costly) or 'dominated' (less effective, more costly). It is not possible to tell this from the ICER itself. Therefore, it is clearer to state that the intervention is dominant or dominated, rather than to present ICERs. We report the proportion of ICER estimates in each of the four quadrants of the cost-effectiveness plane. We present mean ICERs for all results, indicating for negative ICERs whether the intervention is dominant or dominated.

Cost-effectiveness acceptability curves [48] were constructed to express the probability that PINCER is cost effective as a function of the decision maker's ceiling cost-effectiveness ratio (λ) for base-case, sensitivity and scenario analyses [49].

The incremental net monetary benefit (INB) was estimated using the following formula:

$$\mathrm{INB}\ (\lambda) = \lambda\,(\mathrm{QALY}_{\mathrm{PINCER}} - \mathrm{QALY}_{\mathrm{Simplefeedback}}) - (\mathrm{Cost}_{\mathrm{PINCER}} - \mathrm{Cost}_{\mathrm{Simplefeedback}})$$

Table 3 Probabilities for the 3-month cycle Markov model in the error and non-error groups for each of the six models

Transition probability (alpha, beta)	No error	Error
1 Patients with a medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI		
No GI adverse event → No GI adverse event	0.894*	0.829*
No GI adverse event →GI discomfort**	0.099[26] (15,85)	0.154[26, 37] (15,85)
No GI adverse event →Symptomatic ulcer**	0.0047[26] (98,2)	0.0142[26, 37] (98,2)
No GI adverse event →Serious GI event**	0.0001[26] (99,1)	0.0002[26, 37] (99,1)
No GI adverse event → Death	0.0003[54]	
Discomfort→Discomfort	0.188[26]#	
Discomfort→Symptomatic ulcer	0.0069[26]#	
Discomfort→Serious GI event	0.00015[26]#	
Discomfort→No further GI event	0.802*	
Discomfort→Death	0.0003[54]#	
Symptomatic ulcer→Discomfort	0.148[26]#	
Symptomatic ulcer→Symptomatic ulcer	0.0183[26]#	
Symptomatic ulcer →Serious GI event	0.00039[26]#	
Symptomatic ulcer →No further GI event	0.824*	
Symptomatic ulcer →Death	0.001[55]	
Serious GI event→Discomfort	0.148[26]	
Serious GI event →Symptomatic ulcer	0.0183[26]	
Serious GI event →Serious GI event	0.00039[26]	
Serious GI event→No further GI event	0.725*	
Serious GI event →Death	0.1083[56]	
2 Patients with a history of asthma who have been prescribed a beta-blocker		
No symptoms→No symptoms	0.982*	0.955*
No symptoms→Moderate exacerbation	0.008[28](3,97)	0.031[57] (3,97)
No symptoms→Severe exacerbation	0.002[28] (7,93)	0.007[19] (1,99)
No symptoms→Death	0.008[28] (0.8,99.2)	
Severe exacerbation→No symptoms post event	0.664*	
Moderate exacerbation→Moderate exacerbation	0.111[28] (11,89)	
Moderate exacerbation→Severe Exacerbation	0.111[28] (11,89)	
Moderate exacerbation→Death	0.114[28] (11,89)	
Severe exacerbation→No symptoms post event	0.797*	

Table 3 continued

Severe exacerbation→Death	0.203 [28] (20,80)	
No symptoms post exacerbation→Death	0.008[28] (0.8,99.2)	
3 Patients aged 75 years and older who have been prescribed an ACEI long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months		
No symptoms →No symptoms	0.979*	0.988*
No symptoms→ hyperkalaemia	0.016[23]	0.008 (1,99)[34]
No symptoms→ARF	0.0010 (2,98) [39] [40]	0.0005(1,99)[38]
Hyperkalaemia→ Post Hyperkalaemia	0.996*	
ARF→Post ARF	0.974*	
No symptoms→Dead	0.004 (1,99) [58] [59]	
Hyperkalaemia →Dead	0.004[23]	
Post hyperkalaemia →Dead	0.004[23]	
ARF→Dead	0.026 (2.57, 97.43)[41]	
Post ARF→Dead	0.026 (2.57, 97.43) [41]	
1 Patients receiving methotrexate for at least three months who have not had a recorded full blood count and/or liver function test within the previous three months		
No symptoms →No symptoms	0.9474*	0.9686*
No symptoms→Liver Toxicity	0.0434(4.3,95.7)[60]	0.0228 (2.2, 97.8)[61]
No symptoms→BMS	0.0038(0.38, 99.62,)[62]	0.0032 (0.32,99.68)[61]
BMS → Post BMS	0.9270*	
No symptoms→Dead	0.0054(0.54,99.46)[63]	
Liver Toxicity→Post Liver Toxicity	0.9946*	
Liver Toxicity→Dead	0.098(9.8,91.2)[64]	
BMS→Dead	0.0730(7.3,92.7)[65]	
Post-BMS →Dead	0.0054(7.3,92.7)[63]	
5 Patients receiving lithium for at least three months who have not had a recorded check of their lithium levels within the previous three months		
Supra or therapeutic → Supra or therapeutic	0.6089*	0.7113*
Supra or therapeutic → Sub-therapeutic	0.2463[66] (931.67,2830.33)	0.1440[67] (544.61,3237.39)
Subtherapeutic → subtherapeutic	0.8147*	
Subtherapeutic → relapse: manic	0.0725[46](8.41,136.59)	
Subtherapeutic → relapse: depressed	0.1091[46] (11.67,113.93)	
Subtherapeutic → suicide/dead	0.0037[54, 68] (0.17,218.83)	

Table 3 continued

Supra or therapeutic → relapse: manic	0.0427[46] (10.02, 134.99)	
Supra or therapeutic → relapse: depressed	0.0987[46] (4.95, 140.05)	
Supra or therapeutic → suicide/dead	0.0034[54, 68] (0.10,218.90)	
Relapse: manic → sub-therapeutic	0.1440[67] (544.61,3237.39)	
Relapse: manic → supra or therapeutic	0.8530*	
Relapse: manic → suicide/dead	0.0030[54] (2951,997648)	
Relapse: depressed → sub-therapeutic	0.1440[67] (544.61,3237.39)	
Relapse: depressed → supra or therapeutic	0.8530*	
Relapse: depressed → suicide/dead	0.0030[54]	
6 Patients receiving amiodarone for at least six months who have not had a thyroid function test within the previous six months		
AIT untreated → AIT surgical management	0.0081[23]	0.081[42] (25,67)
AIT untreated → AIT medical management	0.0988[23]	0.9879*
AIT untreated → AIT untreated	0.8964*	0[23]
AIH untreated → AIH medical management	0.0995[23]	0.9945*
AIH untreated → AIH untreated	0·8950*	0[23]
No Symptoms → No Symptoms	0.9622*	
No Symptoms → AIT untreated	0.0233[69] (19,73)	
No Symptoms → AIH untreated	0.0115[69] (14,78)	
No Symptoms → Death	0.0035[54, 70] (3,31)	
AIT untreated → Death	0.0550 [54, 70, 71] (1,11)	
AIT surgical management → Post treated AIT	0.9083*	
AIT surgical management → Death	0.0917[54, 70, 72] (2,17)	
AIT medical management → Post treated AIT	0.9965*	
AIT medical management → Death	0.0035[54, 70] (3,31)	
Post treated AIT → Post treated AIT	0.9965*	
Post treated AIT → Death	0.0035[54, 70] (3,31)	
AIH untreated → Death	0.0055 [54, 70, 72] (1,18)	
AIH treated → AIH treated	0.9965**	
AIH treated → Death	0.0035[54, 70] (3,31)	

[a] Net of other probabilities at this node

[b] Probabilities in non-error model derived from error probabilities multiplied by risk reduction from Pettit et al. [37]

[c] Probabilities are calculated as the probability of no GI adverse event to the respective destination state multiplied by relative risk derived from Pettitt et al. [37]

ACEI ACE inhibitor, *AIH* amiodarone-induced hypothyroidism, *AIT* amiodarone-induced thyrotoxicity, *ARF* acute renal failure, *BMS* bone marrow suppression, *GI* gastrointestinal, *PPI* proton pump inhibitor

Table 4 Utilities and costs for health states in the error and non-error groups for each of the six models

Health state	Utility weights (SE, alpha, beta where available)	Mean cost per patient [£; year 2012 values] (SE, alpha, beta)
1. Patients with a medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI		
No GI adverse events	0.85 [73] (0.03, 119.57, 21.10)	8 (0.4, 400, 0.02)
Discomfort[c]	0.009 [43] (0.0018, 2,500, 0.000036)	175 (8, 400, 0.4)
Symptomatic ulcer[c]	0.130 [43] (0.0026, 2,500, 0.000052)	756 (38, 400, 1.9)
Serious GI event[c]	0.180 [43] (0.0036, 2,500, 0.000072)	10,171 (509, 400, 25.4)
No further GI event following initial GI event[c]	0.85 [73]	143 (7, 400, 0.36)
Death	0	1,685 (85, 400, 4.2)
2. Patients with a history of asthma who have been prescribed a β-blocker		
No symptoms	0.73 (0.03, 159.14, 58.86) [27]	7.7 (1.5, 400, 0.02) [27, 28, 74]
No symptoms post-event	0.73 (0.03, 159.14, 58.86)	5.5 (1.1, 400, 0.01)
Moderate exacerbation	0.67 (0.02, 369.67, 182.08)	81.8 (16.4, 400, 0.2)
Severe exacerbation	0.66 (0.04, 91.91, 47.35)	4,396 (879, 400, 11)
Death	0	0
Death after severe exacerbation	0	1,291 (65, 400, 3.2)
3. Patients aged 75 years and older who have been prescribed an ACEI long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months		
No symptoms	0.78 (SE 0.013, 95 % CI 0.75–0.81) [75]	42 (2, 400, 0.1) [53, 76]
Hyperkalaemia	0.60 (no range reported) [77]	1,569 (78, 400, 3.9) [78]
ARF	0.44 (0.032, 0.38, 0.50) [79]	3,226 (161, 400, 0.3) [78]
No hyperkalaemia post-hyperkalaemia	0.73 (0.019, 0.69, 0.77) [80]	125 (6, 400, 8.06) [53, 76]
No ARF post-ARF	0.60 (no range reported) [81]	125 (6, 400, 0.3) [53, 76]
Death	0	0
4. Patients receiving methotrexate for at least 3 months who have not had a recorded full blood count and/or liver function test within the previous 3 months		
No symptoms	0.90 (SE 0.02, 95 % CI 0.86–0.94) [82]	42 (2, 400, 0.1) [53, 83]
Liver toxicity	0.76 (0.02, 0.72, 0.80) [84]	2,620 (131, 400, 6.6) [78]
BMS	0.75 [23]	2,943 (147, 400, 7.4) [78]
No liver toxicity post-liver toxicity	0.84 [23]	125 (6, 400, 0.3) [53, 83]
No BMS post-BMS	0.80 [23]	125 (6, 400, 0.3) [53, 83]
Death	0	0
5. Patients receiving lithium for at least 3 months who have not had a recorded check of their lithium levels within the previous 3 months		
Stable: subtherapeutic	0.74 (0.23, 1.95, 0.69) [44]	0 (error)
		8 (0.4, 400, 0.02) (no error)
Stable: supra/therapeutic	0.71 (0.22, 2.31, 0.94) [44]	133 (7, 400, 0.33) (error)
		141 (7, 400, 0.3) (no error)
Mania (OP) mild relapse[a]	0.56 (0.27, 1.33, 1.05) [44]	6,026 (301, 400, 15.07)
Mania (IP) mild relapse[a]	0.26 (0.29, 0.33, 0.95) [44]	10,093 (595, 400, 25.23)
Mania (OP) moderate relapse[a]	0.54 (0.26, 1.44, 1.23) [44]	6,026 (301, 400, 15.07)
Mania (IP) moderate relapse[a]	0.23 (0.29, 0.25, 0.85) [44]	10,093 (595, 400, 25.23)
Depression (OP) mild relapse[b]	0.70 (0.20, 2.98, 1.28) [45]	7,496 (375, 400, 18.74)
Depression (IP) mild relapse[b]	0.33 (0.36, 0.22, 0.45) [45]	10,093 (595, 400, 25.23)
Depression (OP) moderate relapse[b]	0.63 (0.19, 3.44, 2.02) [45]	7,496 (375, 400, 18.74)
Depression (IP) moderate relapse[b]	0.27 (0.34, 0.19, 0.52) [45]	10,093 (595, 400, 18.74)
Death	0	284 (14, 400, 0.71)
6. Patients receiving amiodarone for at least 6 months who have not had a thyroid function test within the previous 6 months		
No symptoms	0.78 (0.21, 78, 22) [85]	Error 97 (5, 400, 0.24)
		No error 99 (5, 400, 0.25)

Table 4 continued

Health state	Utility weights (SE, alpha, beta where available)	Mean cost per patient [£; year 2012 values] (SE, alpha, beta)
Untreated AIH	0.60 (0.21, 82.1, 17.9) [86]	Error 97 (5, 400, 0.24)
		No error 99 (5, 400, 0.25)
Treated AIH	0.65 (0.21, 85.77, 14.23) [87]	160 (8, 400, 0.40)
Untreated AIT	0.58 (0.21, 80, 20) [86]	Error 152 (8, 400, 0.38)
		No error 154 (8, 400, 0.38)
Medically treated AIT	0.76 (0.21, 97.55, 2.45) [87]	360 (18, 400, 0.90)
Surgically treated AIT	0.73 (0.21, 95, 5) [88]	3,210 (160, 400, 8.02)
Post-treated AIT	0.76 (0.21, 97.55, 2.45) [87]	101 (5, 400, 0.25)
Death	0	0

ACEI ACE inhibitor, *AIH* amiodarone-induced hypothyroidism, *AIT* amiodarone-induced thyrotoxicity, *ARF* acute renal failure, *BMS* bone marrow suppression, *GI* gastrointestinal, *IP* inpatient, *OP* outpatient, *SE* standard error

[a] 20 % patients treated as OP, 80 % patients treated as IP

[b] 90 % patients treated as OP, 10 % patients treated as IP

[c] Decrements from Spiegel et al. [43], assuming an SE of 2 % around these numbers and applying a gamma distribution

where it was calculated for λ ranging from £0 to £160,000.

2.12 Sensitivity and Scenario Analysis

The errors included in the intervention were varied in the following scenario analyses: each error separately; three PINCER primary outcome errors only; prescribing errors only; monitoring errors only; dominated errors removed; and only the two most cost-effective errors included. The costs associated with the PINCER intervention were varied. The practice size affects the intervention cost and some costs are practice level (i.e. costs per patient would be lower for larger practices), so this was varied.

3 Results

Table 5 summarises the 5-year costs and outcomes derived from each error-specific model.

Table 6 summarises the inputs, QALYs gained and costs for the deterministic incremental analysis of the PINCER intervention and simple feedback practices. The PINCER intervention generated 0.81 extra QALYs per practice, compared with simple feedback at £2,679 less per practice. Therefore, the PINCER intervention 'dominated' simple feedback.

In the probabilistic analysis, mean [standard error (SE)] QALYs generated per simple feedback and PINCER practice were 3.141 (0.003) and 3.142 (0.003), respectively. Costs (SE) incurred per simple feedback and PINCER practice were £853 (3.05) and £844 (3.07), respectively. The PINCER intervention generated 0.001 extra QALYs per practice compared with simple feedback, at £4.20 less per practice. Despite this extremely small set of differences in costs and outcomes, the PINCER intervention dominated simple feedback with a mean ICER of −£3,936 (SE £2,970) (Table 7; Fig. 2).

At the National Institute of Health and Care Excellence (NICE) unofficial threshold, ceiling willingness-to-pay for a QALY of £20,000, PINCER reached a 59 % probability of being cost effective. The probability of PINCER being cost effective did not increase beyond this percentage (see Fig. 3). The net benefit statistic, applying

Table 5 Summary of key cost and outcome parameters derived from each outcome measure-specific model

Error	QALYs (SE) generated per patient		Cost [£] (SE) per patient	
	Error	Non-error	Error	Non-error
NSAID	3.89 (0.002)	3.89 (0.002)	2,517 (200)	2,118 (308)
β-Blocker	2.90 (0.23)	3.00 (0.26)	771 (410)	338 (255)
ACE inhibitor	3.40 (0.28)	3.43 (0.29)	1,809 (501)	1,554 (504)
Methotrexate	3.83 (0.24)	3.92 (0.26)	2,513 (571)	1,865 (544)
Lithium	3.05 (0.60)	3.05 (0.60)	13,005 (1,585)	12,410 (1,716)
Amiodarone	3.43 (0.001)	3.51 (0.001)	1,953 (260)	1,990 (252)

Table 6 Summary of inputs and incremental cost-effectiveness ratios generated for deterministic incremental analysis of PINCER intervention versus simple feedback

Error	Prevalence of patient group in practice [15]	Simple feedback event rate per practice	RRR PINCER	QALYs generated per practice[a]		Cost (£) per practice		QALY difference per practice	Cost difference per practice (£)
				Simple feedback	PINCER	Simple feedback	PINCER		
NSAID	7 %	0.04	0.35	256.6	256.6	100,968	100,635	0.01	−333
β-Blocker	71 %	0.03	0.17	1,530.3	1,530.5	256,226	255,205	0.26	−1,022
ACE inhibitor	16 %	0.08	0.36	407.6	407.8	119,066	117,742	0.16	−1,324
Methotrexate	4 %	0.31	0.19	124.6	124.8	57,020	55,991	0.16	−1,028
Lithium	1 %	0.40	0.11	24.2	24.2	100,857	100,636	0.00	−220
Amiodarone	1 %	0.45	0.25	36.8	37.1	16,788	17,023	0.21	234
Difference in intervention cost/practice									1,014
Total								0.81	−2,679
ICER								(−£3,037) dominant	

ICER incremental cost-effectiveness ratio, *RRR* relative risk reduction

[a] QALYs and cost per practices are calculated for a practice with a population at risk of the six errors of 799 patients

Table 7 Mean (SE) ICERs, percentage of ICERs in each quadrant of the cost-effectiveness plane, probability of cost effectiveness at $\lambda < £20,000$ for base case, sensitivity and scenario analyses

	Mean (SE) ICER (£/QALY)[a]	Percentage of ICERs in each quadrant				Probability of cost effectiveness at $\lambda < £2,0000$ (%)
		NE	SE	SW	NW	
Base case	−3,936 (2,970) DOMINANT	20	40	25	15	59
NSAIDs only	−22,055 (108) DOMINANT	1	99	0	0	99
β-Blocker only	2,610 (3,691)	31	34	18	18	64
ACEI only	−7,207 (5,685) DOMINATED	42	1	1	56	35
Methotrexate only	4,960 (8,760)	9	54	25	11	67
Lithium only	−27,577 (21,995) DOMINANT	12	38	34	15	63
Amiodarone only	515 (16)	68	32	0	0	100
Primary errors only	−1,515 (2,768) DOMINANT	46	6	4	45	46
Monitoring errors only	−36,701 (97,550) DOMINANT	21	36	26	18	54
Prescribing errors only	−1,057 (1,025) DOMINANT	34	23	13	30	63
Base case minus ACEI	−2,595 (3,617) DOMINANT	11	56	26	8	65
NSAIDS and amiodarone only	−902 (20) DOMINANT	25	75	0	0	100
Reduction in intervention costs (all 6 errors)						
−10 %	−3,898 (2,948) DOMINANT	20	40	25	15	59
−30 %	−3820 (2,907) DOMINANT	20	40	25	15	60
−50 %	−3,743 (2,869) DOMINANT	19	41	26	14	60
Number of patients at risk per practice (proxy for practice size), base case $n = 799$ (all 6 errors)						
600	−4064 (3,046) DOMINANT	21	39	24	16	59
1,000	−3,859 (2,927) DOMINANT	20	40	25	15	59
2,000	−3,704 (2,649) DOMINANT	19	41	26	14	60

ACEI ACE inhibitors, *ICER* incremental cost-effectiveness ratio, *SE* standard error

[a] Negative ICERs can indicate intervention is either dominant (mean ICER in SE quadrant) or dominated (mean ICER in NW quadrant)

a $\lambda = £20,000$, generated a mean of £16 (standard deviation £121; median £22; 2.5th percentile −£218; 97.5th percentile £242).

If the PINCER intervention targeted one of the errors only, the mean (SE) costs per QALY generated were as follows: NSAIDs: −£22,055 (£108), dominant; β-blockers: £2,610 (£3,691); ACEI: −£7,207 (£5,685), dominated; methotrexate: £4,960 (£8,760); lithium: −£27,577 (£21,995), dominant; and amiodarone: £515 (£16). As ACEI was dominated, we re-ran the model without it. This increased the probability of cost effectiveness at $\lambda = £20{,}000$ to 65 %. Including only the two most robust models (NSAIDs and amiodarone) increased the probability of cost effectiveness at $\lambda = £20{,}000$ to 100 %.

Other scenario analyses are reported in Table 7. Varying the cost of the intervention or the practice size had a negligible effect on results (Table 7).

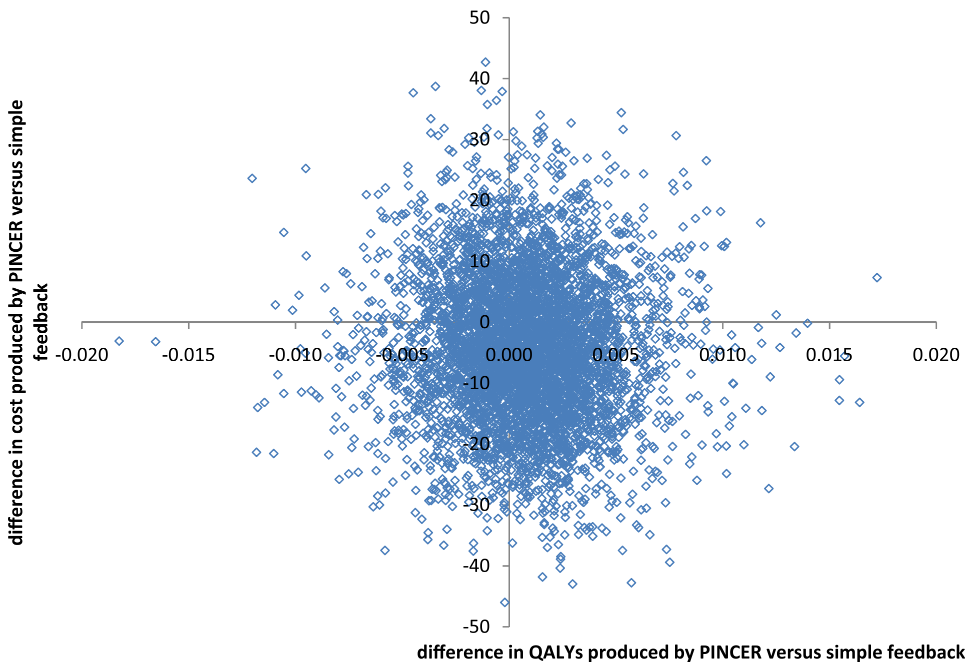

Fig. 2 Cost-effectiveness plane for PINCER intervention versus simple feedback

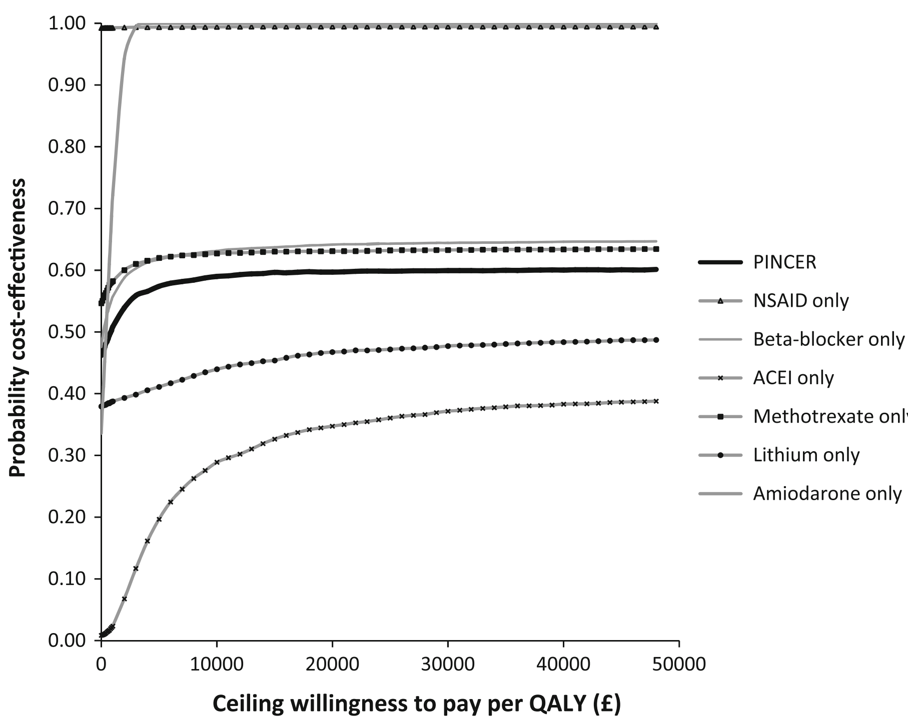

Fig. 3 Cost effectiveness acceptability curve for PINCER intervention versus simple feedback. This graph demonstrates the probability of cost effectiveness at a range of decision-maker ceiling willingness-to-pay values for the PINCER intervention overall, and also when only one error is considered at a time. *ACEI* ACE inhibitor

4 Discussion

PINCER produced a marginal health gain at a slightly reduced overall cost. The results suggest that PINCER increased health gain at a cost per QALY well below most accepted thresholds for technology implementation, usually about £20,000 to £30,000 in the UK [50]. However, results are uncertain due to the poor quality of data to inform the effect of avoiding errors. The wide range around the point estimates of cost effectiveness reflects the uncertainty in some of the individual error models. This uncertainty translates into the probability of cost effectiveness never reaching 90 % and the net benefit statistic, whilst having a positive mean, having a range that incorporates both positive and negative values, suggesting the possibility of both net benefit and net cost.

Targeting NSAID prescribing and amiodarone monitoring errors were the most cost-effective activities. Targeting errors with evidence of effects on patient outcomes will increase the cost effectiveness of the intervention. In contrast, varying the cost of the intervention or the practice size had a negligible effect on results.

Investigation of how each outcome contributes to the cost effectiveness of PINCER demonstrates that correcting errors in NSAID prescribing alone and amiodarone monitoring alone would generate 95 % probabilities of PINCER being cost effective at £10,000 and £0 per QALY gained, respectively. However, correcting errors in β-blocker prescribing and ACEI, diuretic, lithium and methotrexate monitoring may not be cost effective within current thresholds for cost effectiveness. Because NSAID prescribing and amiodarone monitoring accounted for only 8 % of the overall errors corrected, the effects are swamped by the other errors. The quality of the evidence for the clinical and economic impact of NSAID prescribing and amiodarone monitoring errors was better than that available for other errors. The errors with a poor level of evidence showed more uncertainty around their clinical and economic impact within the PINCER intervention.

4.1 Strengths and Limitations

This analysis has included the costs or outcomes that may have been incurred as a result of the errors, giving an estimate of clinical and economic impact of the intervention. We believe that moving beyond the use of process indicators such as error rates and combining multiple error-specific models determining the full economic impact of error-reducing interventions is an important development.

The key limitation of this analysis is the paucity of data upon which to base the estimates of the economic impact of the individual errors. This economic model has been in development since 2008 and was submitted to the funder in 2012. Updating the searches from 2008 to 2010 uncovered no new evidence to populate models as there is very little work going on in this area. One of the limitations of this approach is that the building of each model and incorporation into the composite model is resource intensive. Further work is needed to quantify the actual clinical and economic effect of prescribing and monitoring errors, to provide better data to populate the models. For example, apart from the NSAIDs model, we were not able to incorporate how long a patient might have been exposed to a potential drug interaction or lack of monitoring due to the paucity of data available. Analysis of clinical databases might help us to estimate more accurately the costs and outcomes of errors.

The costs of the simple feedback and PINCER intervention arms reflect one way in which the interventions would be implemented in practice [15]. Differing practice characteristics and methods of service provision may affect costs, although in the cluster RCT there was no evidence of statistically significant interactions between treatment arm and either practice size or practice deprivation for any of the primary outcome measures or intervention costs [21]. This economic analysis did not include any practice costs associated with time spent dealing with errors. It is not clear which arm this omission would favour as practices in the intervention arm spend time with the PINCER pharmacist, but simple feedback practices would have to sort out errors themselves, rather than the pharmacist doing it. However, this means that the costs presented are an underestimate of the real cost to the practice. Ongoing pharmacist support might be needed to maintain (or further reduce) error levels in the intervention group, which would increase intervention costs. We have not considered the costs of implementing this intervention more widely, which would further increase costs.

PINCER was compared with simple feedback rather than usual care because it would have meant identifying patients at risk but not making these known to the practice, which was considered unethical. It may be reasonable to suppose that simple feedback is more effective than usual care. However, this is not the case in another similar study [51] and in the PINCER trial any improvements in the simple feedback error rate were attributed to secular changes [21].

This analysis may have underestimated the true impact of PINCER because not all benefits from a pharmacist intervention may be captured in our analyses. For example, there may have been a reduction in other errors, as the PINCER practice-level approach process of examining causes of errors within a practice may lead prescribers to question other aspects of their prescribing and monitoring.

Some benefits that may be associated with the reduction of medication errors might not contribute to QALYs gained, but may 'go beyond health' [52] and generate other 'outcomes' such as increased patient trust in the NHS associated with lower error rates, regardless of their clinical significance.

4.2 Implications for Policy and Practice

To facilitate formal commissioning decisions under current NHS frameworks, we have attempted to determine the expected cost per QALY gained through the implementation of PINCER. To our knowledge, this is the first attempt to determine the true economic impact of reducing medication errors through the implementation of a complex intervention.

In summary, the results of this study suggest that targeting some errors could prove more cost effective than targeting all errors, and could be considered by policy makers. More research is required to assess costs of wider implementation and to better characterise the impact of individual errors.

5 Conclusions

Despite a significant reduction in errors, the findings of this economic analysis suggest that the PINCER intervention produced marginal health gain at slightly reduced overall cost. We have been unable to quantify with any level of certainty the impact of the PINCER intervention due to the poor quality of data to inform the effect of avoiding errors. Targeting NSAID prescribing and amiodarone monitoring errors were the most cost-effective activities. The wide range around this ICER reflects the uncertainty around the real effect of some errors. Targeting errors with evidence of effects on patient outcomes will increase the cost effectiveness of the intervention. More evidence is critical to support the selection of evidence-based errors for this type of intervention.

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Conflicts of interest The authors (Rachel A. Elliott, Koen Putman, Matthew Franklin, Nick Verhaeghe, Lieven Annemans, Martin Eden, Jasdeep Hayre, Aziz Sheikh, Sarah Rodgers and Anthony J. Avery) declare that they have no competing interests.

Author contributions Rachel A. Elliott designed and led the economic analysis, including all error models, intervention costs and composite model, led drafting of the manuscript and was on the Trial Management Group.

Koen Putman designed the NSAIDs and β-blockers model, contributed to all other models, built the composite model and was involved in the drafting of the manuscript.

Matthew Franklin designed the lithium model and was involved in the drafting of the manuscript.

Nick Verhaeghe designed the methotrexate and ACEI models and was involved in the drafting of the manuscript.

Lieven Annemans contributed to the design of the composite model and was involved in the drafting of the manuscript.

Martin Eden contributed to intervention cost analysis, CHEERS criteria and was involved in the drafting of the manuscript.

Jasdeep Hayre designed the amiodarone model and was involved in the drafting of the manuscript.

Sarah Rodgers was the trial coordinator from the start of the trial to June 2009 and was involved in the drafting of the manuscript.

Aziz Sheikh contributed to the design and execution of the PINCER trial and was involved in the drafting of the manuscript.

Anthony J. Avery was the principal investigator, had overall responsibility for the day-to-day management of the trial and for the conduct of the trial in the area around Nottingham, and was involved in the drafting of the manuscript.

Rachel A. Elliott will act as overall guarantor.

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